Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 22: Assume, Then Derive

Direct Proof

Chartrand & Zhang, Section 3.2

Name: _____ Date: _____

A new unit — and the centre of the course

You have written proofs already: double inclusion in Week 5, equivalence relations in Week 6, bijections in Week 7. What you have not had is the *method* named and practised on its own. That starts now, and the next three weeks are the heart of CS 341.

By the end of today you should be able to

1. State the definitions of even and odd precisely.
2. Write a direct proof of an implication about integers.
3. Explain why "$2b - 1$" does not establish oddness.

1. The template

Almost every theorem you will prove has the form

$$\forall x \in S,\ P(x) \Rightarrow Q(x).$$

Direct proof

Assume $P(x)$ is true for an **arbitrary** $x \in S$. Derive $Q(x)$.
That is all. The implication is false in only one row — P true and Q false — so ruling out that row proves it.

The word *arbitrary* is doing the work: you may assume $x \in S$ and $P(x)$, and nothing else.

2. Definitions must be exact

Definition 3.15

An integer n is **even** if $n = 2a$ for some integer a.
An integer n is **odd** if $n = 2b + 1$ for some integer b.

Say "for some integer"

Both definitions end with *for some integer*. If you write $n = 2a$ without saying a is an integer, you have said nothing — every real number is twice something.
The last line of every parity proof depends on this phrase. Omitting it is the most common way these proofs lose points.

We are also allowed to assume (3.14): if a and b are integers, so are $-a$, $a + b$, and ab.

We may *not* assume things like "the sum of two evens is even." Those must be proved — and once proved, may be used.

3. Worked: Result 3.16

Result 3.16

If n is an odd integer, then $5n + 3$ is an even integer.

Proof. Let n be an odd integer. Then $n = 2k + 1$ for some integer k. Then

$$5n + 3 = 5(2k + 1) + 3 = 10k + 5 + 3 = 10k + 8 = 2\big(5k + 4\big).$$

Since $5k + 4$ is an integer, $5n + 3$ is even. ■

The shape of every one of these

1. Assume the hypothesis and **write it as an equation** using the definition.
2. Substitute into the expression you care about.
3. Algebra, until the target form appears.
4. **Factor out the 2** and say the remaining factor is an integer.

Step 4 is where the proof actually finishes. Everything before it is arithmetic.

4. Worked: Result 3.17, and a trap

Result 3.17

If n is an even integer, then $n^2 + 4n - 3$ is odd.

Proof. Let n be even, so $n = 2k$ for some integer k. Then

$$n^2 + 4n - 3 = 4k^2 + 8k - 3.$$

Now we must reach the form $2b + 1$. Note

$$4k^2 + 8k - 3 = 4k^2 + 8k - 4 + 1 = 2\big(2k^2 + 4k - 2\big) + 1.$$

Since that is an integer, $n^2 + 4n - 3$ is odd. ■

Why not stop at $2(2k^2 + 4k - 1) - 1$?

You could also write $4k^2 + 8k - 3 = 2(2k^2 + 4k - 1) - 1$. That is true — and it proves nothing. The definition says odd means $2b + \mathbf{1}$, not $2b - 1$. The two describe the same integers, but a proof must match the *definition as written*. Adjust the algebra until it does.
Chartrand flags this explicitly. It is the single most common place these proofs go wrong.

5. Beyond parity

Result 3.19

If r and s are rational numbers, then $r + s$ is rational.

Recall r is **rational** if $r = a/b$ with $a, b \in \mathbb{Z}$ and $b \neq 0$.

Proof. Let $r = a/b$ and $s = c/d$ with $a, b, c, d \in \mathbb{Z}$ and $b, d \neq 0$. Then

$$r + s = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

Since $ad + bc$ and bd are integers and $bd \neq 0$, the sum $r + s$ is rational. ■

Same four steps: unfold the definition, substitute, do algebra, check the target form.

Exit ticket

1. Prove: if n is odd, then $7n + 4$ is odd.

2. Why is "$n = 2a$" insufficient without saying a is an integer?

Before Friday

Read Section 3.3, "Proof by Contrapositive," pp. 96–98.

Self-check (not collected). Section 3.2, Exercises 1, 3, 5, and 7 — odd-numbered, answers in the back.

Homework 8 is due Friday, October 9, 11:59 p.m.

Questions I still have